

CRYPTO LIQUIDITY AND RISK APPETITE

EVIDENCE FROM STABLECOIN ISSUANCE AND OTHER
INDICATORS

WHY STABLECOINS MAY MATTER BEYOND CRYPTO

- Core settlement asset in crypto markets
- Used for trading, collateral, and capital reallocation
- Supply and exchange balances proxy crypto liquidity
- Crypto liquidity may reflect broader risk-taking
- Stablecoins may serve as a predictive signal

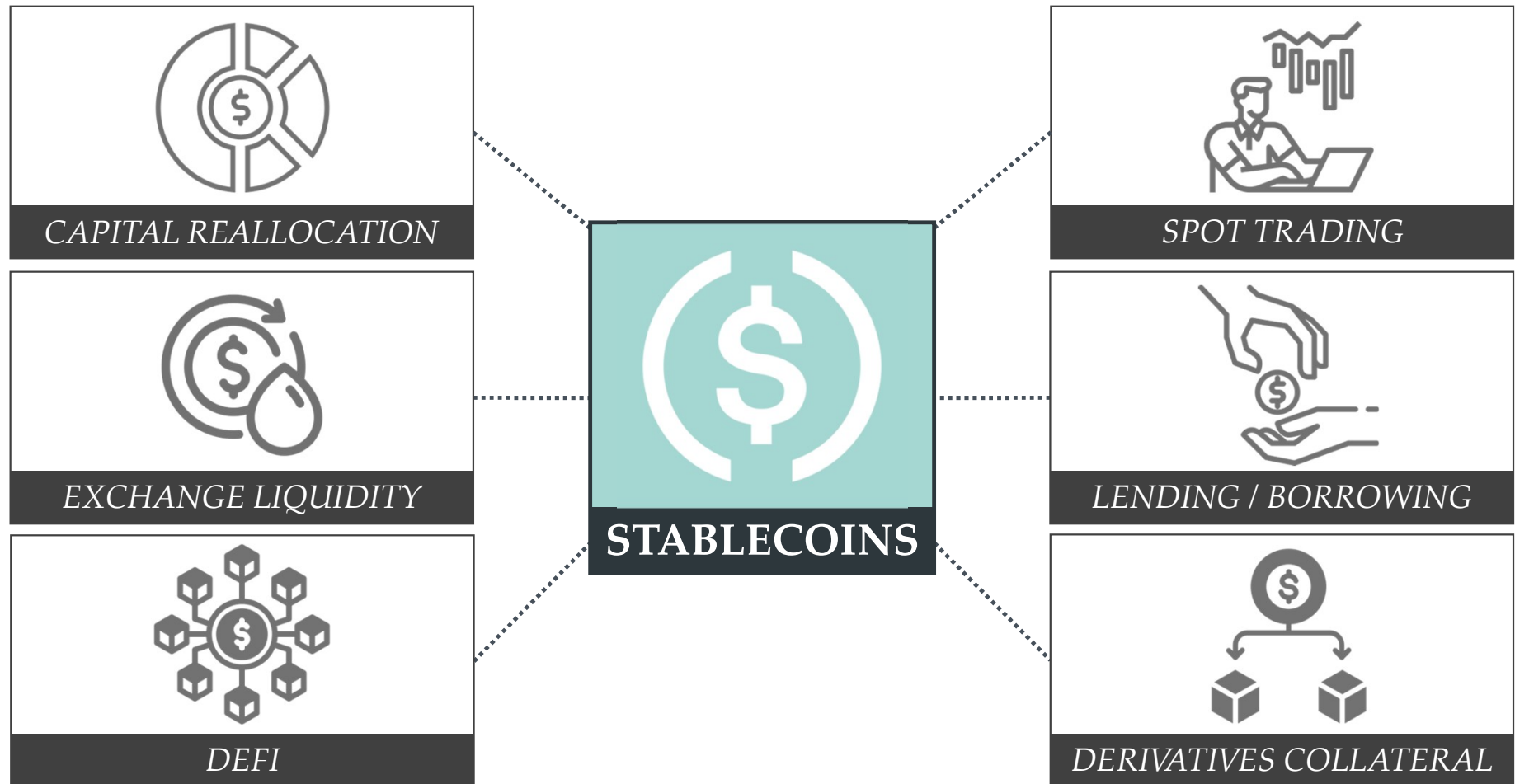


Do crypto liquidity indicators predict global risk appetite?

- Focus on stablecoins, flows, reserves, and market activity
- Risk appetite measured through equities, volatility, and credit spreads
- **Contribution:** links stablecoin liquidity to broader financial conditions
- **Aim:** predictive evidence, not causal identification

Concept	Measured by
Crypto liquidity	Stablecoin supply, netflows, reserves, open interest, funding rates
Global risk appetite	S&P 500, VIX, HY spreads, IG spreads

STABLECOINS AS MARKET LIQUIDITY INFRASTRUCTURE



WHERE THIS PROJECT FITS IN THE LITERATURE

Literature strand	Relevance to project
Stablecoins and crypto liquidity	Stablecoins support trading, settlement, arbitrage, and liquidity provision
Crypto assets and broader markets	Crypto is often speculative, but linkages to traditional markets can vary over time
Predictive finance and sentiment	Tests whether market-based indicators contain information about future risk appetite

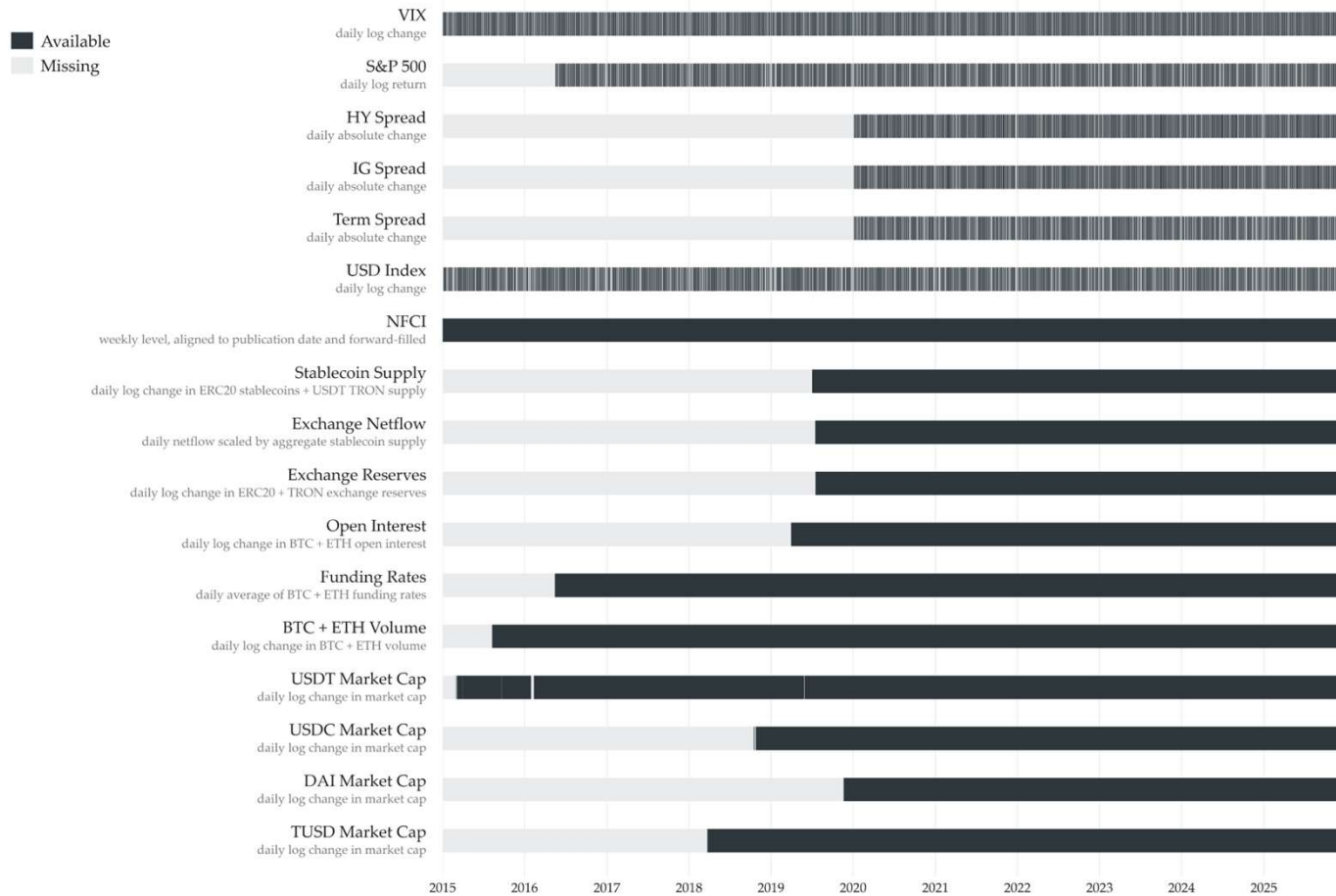
WHAT ENTERS THE EMPIRICAL ANALYSIS?

Variable group	Measures	Role
Global risk appetite	S&P 500, VIX, HY spreads, IG spreads	Dependent variables
Stablecoin liquidity	Market caps, exchange reserves, netflows	Main predictors
Crypto market activity	Open interest, funding rates, BTC/ETH volume	Additional predictors
Financial controls	Term spread, dollar strength, NFCI, lagged dependents	Confounders / controls

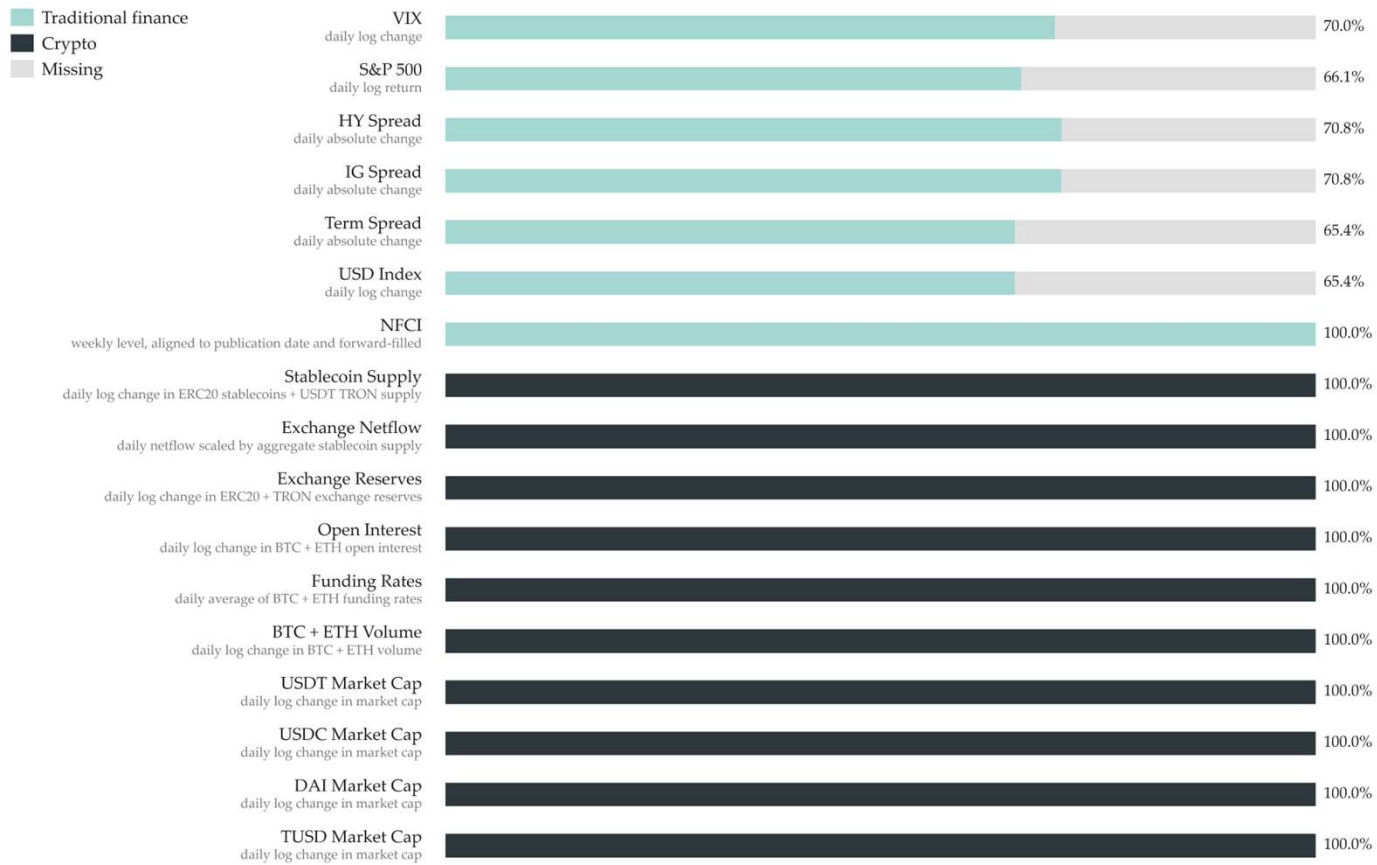
FROM RAW DATA TO ANALYSIS-READY VARIABLES

- All series aligned to daily frequency
- Common sample starts around 2020-01-01
- Transformations ensure comparability over time
- Price and market-cap variables enter as log changes
- Spreads and rate variables enter as daily changes
- Netflows scaled by aggregate stablecoin market cap

DATA AVAILABILITY



DATA AVAILABILITY



METHOD I – TESTING PREDICTIVE CONTENT

- Tests whether crypto liquidity at t predicts risk appetite at $t+h$
- Dependent variables: S&P 500, VIX, HY spreads, IG spreads
- Predictors: stablecoin liquidity and crypto market activity
- Forecast horizons: 1, 5 and 20 days
- Controls added gradually to test robustness
- Interpretation focuses on predictive content, not causality
- HAC standard errors account for overlapping horizons and serial correlation

$$Y_{t+h} = \alpha + \beta X_t + \gamma' C_t + \varepsilon_{t+h}$$

Y_{t+h} risk appetite measure at horizon $t + h$
 X_t crypto liquidity proxy at time t
 C_t control variables at time t
 h forecast horizon



METHOD I – EMPIRICAL FINDINGS

	S&P 500	VIX	HY Spread	IG Spread
Stablecoin Supply	+	-	+	
Exchange Netflow	+	-	-	-
Exchange Reserves	-	+		+
Open Interest		+	-	-
Funding Rate	+		-	-
BTC + ETH Volume				

Dot size = recurrence of significant results across horizons/specifications ($p < 0.10$)
 Dot color = strongest significance in cell: black < 1%, dark < 5%, light < 10%
 +/- = average coefficient direction among significant results
 Signs show raw predicted response. Risk-on = S&P 500 up, VIX down, HY/IG spreads down

Interpretation

- Results suggest that crypto liquidity contains some predictive information
- Signals are most visible for VIX and credit spreads
- Open interest and stablecoin exchange-side appear most informative

Caveats

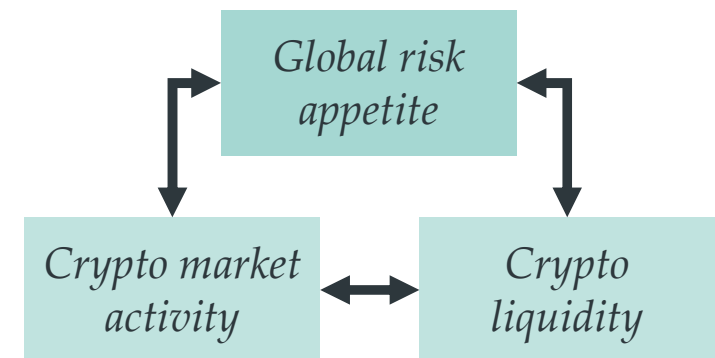
- Predictive association, not causal identification
- Many tested specifications create a multiple-testing concern
- Results may be sample-sensitive due to different data availability
- Statistical significance does not necessarily imply large economic magnitude

METHOD II – DYNAMIC INTERACTION THROUGH VAR

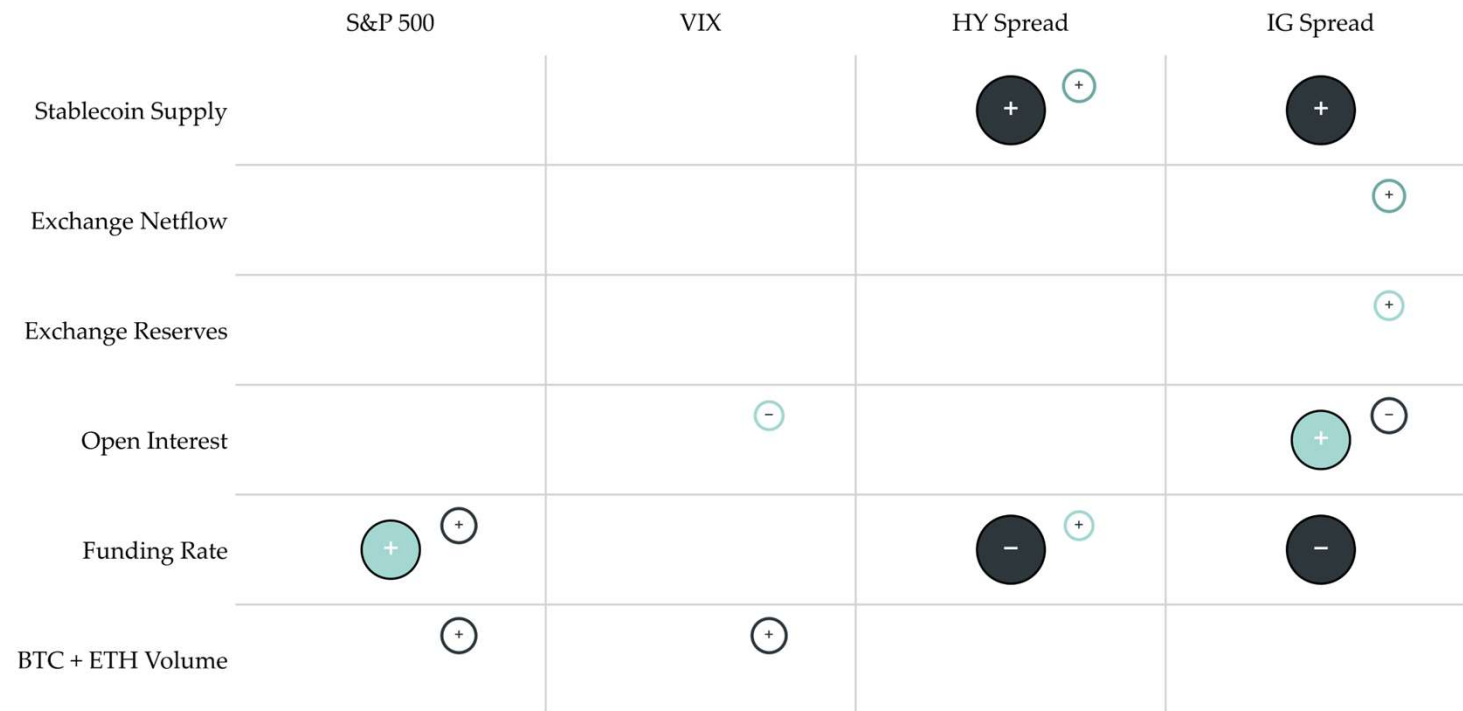
- VAR models variables jointly as an interconnected system
- Captures feedback between crypto liquidity and returns, and risk appetite
- Uses lagged values of all variables to explain current movements
- Granger causality tests whether crypto liquidity improves prediction
- Impulse responses trace effects of liquidity shocks over time
- Complements predictive regressions with a dynamic perspective

$$Y_t = c + A_1 Y_{t-1} + A_2 Y_{t-2} + \dots + A_p Y_{t-p} + u_t$$

Y_t	<i>vector of variables</i>
$A_1, \dots, A_p$	<i>lag coefficient matrices</i>
p	<i>number of lags</i>
u_t	<i>innovations / shocks</i>



METHOD II – EMPIRICAL FINDINGS



Main dot = crypto to risk Granger significance: dark <1%, medium <5%, light <10%
 Main +/- = cumulative risk-variable response after crypto-indicator shock
 Corner circle = risk to crypto Granger significance; +/- = cumulative crypto response after risk-variable shock
 Signs show raw predicted response. Risk-on = S&P 500 up, VIX down, HY/IG spreads down

Interpretation

- Results support dynamic predictive links between crypto and risk appetite
- Evidence is concentrated in credit markets rather than VIX
- Stablecoin supply predicts credit-spread movements
- Funding rates behave more like a speculative sentiment proxy

Caveats

- Granger causality is predictive, not structural causality
- IRFs are point estimates; confidence bands still needed
- Shock magnitudes are not yet standardized
- Multiple tests increase the risk of false positives
- Results may depend on lag choice, sample period, and variable construction

- Strongest cross-method agreement:
 - Funding rates
 - Open interest
- Credit spreads are the most consistently predictable risk variables
- Stablecoin supply shows some agreement, especially for credit spreads
- Exchange netflows / reserves
 - Strong in predictive regressions
 - Weak in crypto → risk VAR tests
- VAR results reveal reverse predictability
 - Suggesting feedback or common-risk-factor dynamics
- BTC + ETH volume remains weak as a forward-looking indicator

(TENTATIVE) ECONOMIC INTERPRETATION

- Crypto indicators contain predictive information about risk conditions
 - Especially credit spreads
- Funding rates and open interest are most consistent with a risk-on/speculative-positioning channel
- Stablecoin variables are more ambiguous
 - Supply / reserve / netflow increases may reflect demand for crypto *cash* during stress
- Weak BTC/ETH volume results suggest raw trading activity is too noisy

Overall: evidence supports a predictive common-risk-factor story, not direct causality

- **Robustness of VAR specification**
Alternative lags, subsamples, and joint crypto-indicator VAR
- **Robustness of variable construction**
NFCI control adjustment and alternative stablecoin measures
- **Stronger IRF interpretation**
Standardized shocks, confidence bands, and cumulative 5/10/20-day responses
- **Common-factor and feedback dynamics**
Further assess bidirectionality and possible shared risk factor

**THANK YOU FOR
YOUR ATTENTION!**

Literature

- Adrian, Boyarchenko & Giannone (2019): Vulnerable Growth
- Auer, Cornelli, Doerr, Frost & Gambacorta (2023): Crypto trading and Bitcoin prices
- Baker & Wurgler (2006): Investor sentiment and the cross-section of stock returns
- Baur, Hong & Lee (2018): Bitcoin: Medium of exchange or speculative asset?
- Campbell & Thompson (2008): Predicting the equity premium out of sample
- Cong, Li & Wang (2021): Tokenomics: Dynamic adoption and valuation
- Corbet, Meegan, Larkin, Lucey & Yarovaya (2018): Dynamic relationships between cryptocurrencies and other financial assets
- Corbet, Hou, Hu, Larkin & Oxley (2020): Cryptocurrency safe-havens during the COVID-19 pandemic
- Griffin & Shams (2020): Is Bitcoin really untethered?
- Lyons & Viswanath-Natraj (2023): What keeps stablecoins stable?
- Makarov & Schoar (2020): Trading and arbitrage in cryptocurrency markets

Data sources

- CBOE: VIX historical data
- FRED: S&P 500, HY spread, IG spread, 10Y–2Y Treasury spread, Broad U.S. Dollar Index
- Chicago Fed: National Financial Conditions Index
- CoinGecko: Stablecoin market caps, BTC/ETH prices and trading volumes
- CryptoQuant: Stablecoin exchange netflows, exchange reserves, BTC/ETH open interest, BTC/ETH funding rates

External assistance

ChatGPT was used for coding support, wording suggestions, and critical feedback on presentation structure. All empirical decisions, interpretation, and final responsibility remain with the author.